A Time-Domain Analogue to Fibonacci Structure via Phase-Gated AR(2) Dynamics: Reply to Boman on Tissue Fibonacci Patterns and Colonic Crypt Renewal

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ABSTRACT
Boman et al. have developed an elegant spatial framework in which simple rules for asymmetric division and niche geometry generate Fibonacci-like cell-number patterns in renewing tissues [1, 2]. Their tissue-renewal model based on age-structured, asymmetric cell division produces Fibonacci sequences in time, and their recent five-rule tissue code for colonic crypts explains how spatial organization is maintained [2]. In this reply, I propose a complementary time-domain analogue: a Phase-Gated Order-2 Autoregressive (PAR(2)) model for crypt renewal in which Fibonacci-like structure appears in the dynamics of phase-aligned observables rather than only in static spatial counts. I outline the minimal mathematics of the PAR(2) construction, show that crypt-relevant circadian genes in intestinal organoids are well-described by stable, complex AR(2) roots, and map the temporal parameters onto Boman's five spatial rules. Finally, I derive testable predictions for tuft cell and deep crypt secretory (DCS) behaviour under circadian disruption, injury, and chronotherapy. The aim is to offer a simple time-domain model that sits naturally alongside Boman's spatial code and invites joint experimental tests.

KEYWORDS
AR(2); autoregressive model; circadian; colonic crypt; Fibonacci; phase-gated; tissue renewal; tuft cells

1. Context: From Spatial Fibonacci Patterns to a Tissue Code

Fibonacci numbers and related sequences appear widely in biology, from phyllotaxis to branching and aperiodic order in tissues. A recent survey by Bormashenko reviews how Fibonacci-like structure can arise from symmetry, scaling, and growth constraints across multiple systems [3].

Boman et al. built on earlier work by Spears and Bicknell-Johnson on age-structured asymmetric division to propose a tissue-renewal model in which mature and immature cells follow simple division rules that yield Fibonacci sequences in time and hierarchical patterns in tissues [1]. In their Fibonacci Quarterly paper, they show that asymmetric cell division with appropriate maturation delays generates Fibonacci cell number sequences and rational ratios of immature to mature cells over time.

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In their recent Biology of the Cell article, Boman and colleagues go further, proposing that a small set of five biological rules—a "tissue code"—governs dynamic organization of cells in colonic epithelium [2]. These rules constrain: where stem cells reside; how, when, and in which direction they divide; how many times progeny may divide; how long differentiated lineages live; and when and where crypt fission occurs. Agent-based simulations driven by these rules quantitatively match real crypt architecture and suggest that the same code may generalize to other tissues [2].

Complementing this, Nguyen, Lausten and Boman provide a comprehensive review of colonic crypt cellular composition and signalling pathways, including detailed discussion of $LGR5^+$ stem cells, transit-amplifying progenitors, colonocytes, goblet cells, Paneth-like deep crypt secretory cells (DCS), enteroendocrine cells, M cells and tuft cells, together with WNT, Notch, BMP and FGF signalling and their disruption during colorectal cancer and EMT-driven plasticity [6]. This review offers a biologically rich backdrop for the present time-domain model.

Taken together, Boman's work therefore offers: (1) a time-based Fibonacci model for population growth under asymmetric division [1]; (2) a spatial five-rule code that reproduces static tissue organization [2]; and (3) an integrated crypt-level description of cell types and signalling, including tuft and DCS lineages, via Nguyen–Lausten–Boman [6]. This reply asks: what is the simplest time-series model at the crypt level that could implement and complement this spatial code?

2. Aim of This Reply

The main proposal is a Phase-Gated AR(2) (PAR(2)) model, originally developed in a more general form for crypt dynamics in the Integrated Temporal-Spatial Oscillator Hypothesis for Crypt Renewal [11] and applied here specifically to Boman's Fibonacci tissue code:

- The state variable x_t is a phase-aligned crypt-level observable, such as the proportion of tuft cells, a weighted renewal index, or a composite of several stem and niche markers.
- The dynamics obey a second-order autoregression in discrete time: $x_t = \phi_1(\theta_t)\, x_{t-1} + \phi_2(\theta_t)\, x_{t-2} + \varepsilon_t$, where θ_t is a phase variable (e.g., circadian or cell-cycle phase), and the coefficients switch or vary smoothly with phase.
- Under mild constraints on (ϕ_1, ϕ_2), the characteristic roots approximate those of a Fibonacci-like recurrence, so that Boman's spatial Fibonacci structure is mirrored in temporal return dynamics.

The PAR(2) model is deliberately minimal. It is not intended as a fully mechanistic molecular network, but as a compact time-domain analogue of Boman's code, with three goals: (1) to provide a language for stability and overshoot in crypt renewal in terms of AR(2) roots; (2) to relate circadian clock disruption and niche signalling to deviations from Fibonacci-consistent behaviour; and (3) to generate concrete, falsifiable predictions for tuft cell and DCS dynamics that can be tested in organoids and animal models.

3. A Phase-Gated AR(2) Model

3.1. *Baseline AR(2) dynamics*

Consider a scalar time series $\{x_t\}$ indexed in discrete steps (e.g., equally spaced circadian sampling times or aligned cell-cycle checkpoints). A standard AR(2) process is $x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \varepsilon_t$, with innovations ε_t of mean zero and finite variance. The associated characteristic polynomial $r^2 - \phi_1 r - \phi_2 = 0$ has roots $r_{1,2} = \left(\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}\right)/2$. If both roots satisfy $|r_{1,2}| < 1$, the process is stable; complex-conjugate roots with $|r| < 1$ correspond to a damped discrete-time oscillator with angular frequency $\arg(r)$ and decay determined by $|r|$.

3.2. *Phase gating*

To introduce phase structure, let $\theta_t \in [0, 2\pi)$ be a phase variable evolving approximately linearly in time, $\theta_t = \theta_0 + \Omega t \pmod{2\pi}$, with Ω corresponding to a circadian or cell-cycle frequency. A phase-gated AR(2) model takes $x_t = \phi_1(\theta_t)\, x_{t-1} + \phi_2(\theta_t)\, x_{t-2} + \varepsilon_t$, with $\phi_1(\cdot)$ and $\phi_2(\cdot)$ either piecewise constant or smooth functions of phase. A minimal two-gate construction uses $(\phi_1, \phi_2) = (\phi_1^A, \phi_2^A)$ for θ in gate A, and (ϕ_1^B, ϕ_2^B) for θ in gate B, partitioning the cycle (e.g., subjective day vs. night, or G1/S vs. G2/M). Biologically, the gates can encode a proliferative vs. quiescent phase, a high-Wnt vs. high-YAP state in crypt bases [8], or a tuft/DCS recruitment versus maintenance phase conditioned on inflammatory or microbial cues [5, 10].

3.3. *Fibonacci-consistent constraints in time*

Boman's age-structured asymmetric division model yields Fibonacci sequences for cell counts over discrete time steps by imposing simple rules on division and maturation [1]. In the scalar AR(2) setting, exact Fibonacci dynamics arise when $x_n = x_{n-1} + x_{n-2}$, that is, $(\phi_1, \phi_2) = (1, 1)$, whose characteristic roots are the golden ratio $\varphi = (1 + \sqrt{5})/2$ and its conjugate.

In a crypt-level stochastic setting, it is neither necessary nor realistic to demand exact Fibonacci coefficients. Instead, I require that the AR(2) coefficients for a phase-aligned observable x_t lie within a compact, stable region of the parameter plane, that the associated roots are complex with $|r| < 1$ and phases within a narrow range, and that under projection into a reduced parameter space, the coefficient cloud falls near a golden-ratio-consistent manifold associated with Boman-like asymmetric division rules. Boman's Fibonacci structure is treated as an attractor in coefficient space rather than an exact equality.

It is important to note that the Fibonacci point $(\phi_1, \phi_2) = (1, 1)$ lies outside the AR(2) stationarity triangle, with dominant root $|\lambda| = \varphi \approx 1.618 > 1$. Consequently, no covariance-stationary gene expression series can operate at the exact Fibonacci point; all observed gene eigenvalues satisfy $|\lambda| < 1$. The Fibonacci connection is therefore a geometric boundary landmark of the admissible parameter space, not an attainable operating point. The stable twin of the Fibonacci characteristic polynomial is $1/\varphi \approx 0.618$, which lies well inside the stationarity region and is the biologically relevant reference value.

4. Application to Rhythmic Organoid Data

Stokes et al. studied the role of the core clock gene Bmal1 (*Arntl*) in intestinal stem cell signalling and tumour initiation [8]. Using organoids derived from mouse intestine, they showed that loss of Bmal1 increases self-renewal via YAP1-dependent pathways and disrupts Wnt signalling; transcriptomic profiling was deposited as GEO series GSE157357 [9].

Limitation: model–data mismatch. Boman's five-rule tissue code was developed for human large intestinal crypts [2], while Stokes et al. used mouse small intestinal organoids [8]. Both the species difference (human vs. mouse) and the anatomical region difference (large vs. small intestine) are genuine limitations. Colon and small intestine differ in $LGR5^+$ stem cell density, villus structure, crypt length, and cancer susceptibility; and mouse and human crypt architecture, while broadly conserved, are not identical. The AR(2) fits below should therefore be read as a proof-of-concept demonstration of the method on the best available rhythmic organoid dataset, not as a validated quantitative test of Boman's human-colon model. Direct validation against a human large-intestinal, circadian time-series dataset remains an important open objective.

For an illustrative PAR(2) analysis, consider phase-ordered expression of four crypt-relevant genes from GSE157357: *Lgr5* (stem cell marker), *Arntl* (Bmal1; core clock), *Per2* (clock output), and *Axin2* (Wnt pathway readout). Using a simple least-squares AR(2) fit to z-scored log-expression values ordered by circadian phase, I obtain the characteristic roots shown in Table 1.

Table 1. Illustrative AR(2) fits to selected genes in GSE157357 [8, 9]. Complex-conjugate roots with modulus $|r| < 1$ indicate stable, damped oscillations in the phase-ordered signal. Root magnitudes well below 1 correspond to strongly damped behaviour.

Gene	Roots	Max $\lvert r\rvert$	Stability
Lgr5	$0.049 \pm 0.332j$	0.336	Stable
Arntl	$0.623 \pm 0.551j$	$0.832^{\dagger}$	Stable
Per2	$-0.092 \pm 0.513j$	0.521	Stable
Axin2	$-0.008 \pm 0.464j$	0.464	Stable

† *Corrected value.* The submitted v2.0 manuscript reported $|r| = 0.063$ for *Arntl* (roots $0.017 \pm 0.061j$), which is not reproducible from the WT mean-centred GSE157357 data under any normalisation tested; both the exact PAR(2) pipeline and the log+z-score method give $|r| \approx 0.83$. The corrected characteristic roots are $0.623 \pm 0.551j$ ($|r| = 0.832$). Source of the original discrepancy is under investigation; probable data-indexing error in the original analysis.

All four examples yield complex-conjugate roots with $|r| < 1$, indicating that the phase-ordered expression behaves as a damped oscillator around a mean level. Within the PAR(2) framework, these gene-level AR(2) fits are interpreted as projections of a lower-dimensional crypt-level oscillator into different observables. A natural next step would be to move from univariate AR(2) to multivariate VAR(2) models for panels of genes, fit phase-dependent coefficients, and test whether renewal-relevant composites cluster near a Fibonacci-consistent manifold under wild-type conditions and deviate under Bmal1 loss or environmental circadian disruption.

5. Tuft Cells, Deep Crypt Secretory Cells, and Long-Lived Lineages

The PAR(2) framework requires three specific inputs to be testable: (i) a rhythmic time-series dataset from crypt-relevant tissue, (ii) a long-lived cell type whose dynamics accumulate temporal perturbations over many cycles, and (iii) a niche-supporting cell type whose sustained signalling activity could implement the memory kernels α_1 and α_2. The GSE157357 organoid dataset [8, 9] is chosen because it is the only publicly available circadian time-series profiling crypt stem cell behaviour across wild-type and clock-disrupted conditions simultaneously, making it uniquely suited to test the PAR(2) kernel architecture. Tuft cells are chosen because they are among the longest-lived epithelial cells in the crypt (lifespan ≥ 28 days vs. the 3–5 day average turnover), meaning they accumulate the effects of temporal perturbations across many renewal cycles and are therefore the most sensitive available readout of sustained eigenvalue shifts [10]. DCS cells are chosen because their anatomical position at the crypt base—intermingled with LGR5$^+$ stem cells and providing continuous WNT, EGF and Notch ligand supply—makes them the natural physical implementors of the sustained memory kernels α_1 (fast, NOTCH-mediated) and α_2 (slow, WNT/AXIN2-mediated) in the PAR(2) model [6, 7]. Long-lived lineages more generally are those most likely to accumulate perturbations in AR(2) eigenvalues over many cell cycles and therefore provide the most sensitive readout of sustained temporal disruption.

Tuft cells are rare chemosensory epithelial cells with important roles in immunity and disease. Westphalen et al. used lineage tracing to show that a DCLK1$^+$ tuft cell population in the intestine is long-lived, contributes to regeneration after injury, and can serve as a colon cancer-initiating cell following APC mutation and inflammatory insult [10]. Subsequent work has expanded their role in antimicrobial and anti-parasitic defence and in shaping mucosal immunity [5].

Within the colonic crypt, Nguyen, Lausten and Boman highlight tuft cells, M cells and enteroendocrine cells as important contributors to tumour heterogeneity and treatment resistance, particularly via epithelial–mesenchymal transition and cellular plasticity [6]. Tuft cell expansion is noted as one component of tumour–immune crosstalk and microenvironmental disruption in colorectal cancer [6].

DCS cells have been characterised as Reg4$^+$ niche-supporting cells intermingled with LGR5$^+$ stem cells at the crypt base [6, 7]. Circadian control of stem cell niche components, including regulation of Lgr5 expression by BMAL1 and RNA-binding protein MEX3A [4], suggests that DCS cell function may be phase-modulated, consistent with a role as implementors of the PAR(2) slow memory kernel α_2.

From a tissue-code perspective, long-lived tuft cells and related DCS lineages sit at the intersection of division limits and lifespan (rules on how many times and how long cells persist), niche signals (e.g., IL-25, microbial metabolites, Hippo/Wnt balance), and injury responses [5, 6, 10].

Under perturbations known to affect tuft/DCS biology—e.g., helminth infection, chronic inflammation, microbial changes, or deliberate timing interventions [5, 6, 10]—the PAR(2) model predicts systematic shifts in (ϕ_1, ϕ_2) and therefore in root magnitudes and phases.

6. Mapping Boman's Five Rules onto PAR(2) Dynamics

Boman's five rules for colonic crypt organization can be mapped onto PAR(2) parameters at a coarse level. Rule 1 (spatial positioning of stem cells) constrains the effective pool size and interaction network feeding into x_t, narrowing the allowable region of (ϕ_1, ϕ_2) and preventing runaway growth. Rule 2 (order and direction of divisions) specifies directed movement and differentiation of daughters along the crypt axis, appearing in phase-dependent coefficients $\phi_1(\theta)$, $\phi_2(\theta)$ that encode preferred advance directions. Rule 3 (division limits) places an upper bound on contributions from older generations, pushing the system toward low-order autoregressions with $|r| < 1$. Rule 4 (lifespan of differentiated cells) sets how quickly perturbations decay: shorter lifespans map to smaller $|r|$; extended lifespans push $|r|$ toward 1. Rule 5 (crypt fission) corresponds to a branching event, which can be modelled as a change-point where coefficients switch regime or as a stochastic bifurcation when $|r|$ or innovation variance crosses a critical threshold [2].

In this view, the five rules act as constraints that restrict which AR(2) parameter sets are allowed. Fibonacci structures identified by Boman emerge as special cases where the resulting dynamics stay near golden-ratio-consistent manifolds, both in space and time [1, 2].

A critical clarification: the Fibonacci recurrence $x_n = x_{n-1} + x_{n-2}$ corresponds to $(\phi_1, \phi_2) = (1, 1)$, which lies outside the AR(2) stationarity triangle with dominant root $|\lambda| \approx 1.618 > 1$. This means that exact Fibonacci dynamics would be explosive, not stable. Living tissues operate in the heavily damped interior of the stationarity triangle: the observed root magnitudes in Table 1 range from $|r| = 0.336$ (*Lgr5*) to $|r| = 0.832$ (*Arntl*†), far from the explosive boundary. The biological content of the Fibonacci analogy is therefore not that tissue renewal grows like the Fibonacci sequence—it does not—but that homeostatic regulation holds the AR(2) coefficients near the stable twin of the Fibonacci eigenvalue, $1/\varphi \approx 0.618$, which is well inside the admissible region. The five rules of Boman's tissue code are precisely the biological mechanisms that enforce this heavily damped, sub-unit-root regime.

To make the Fibonacci-consistent manifold concrete: it corresponds to the subset of the stable AR(2) stationarity triangle—the biologically admissible parameter space—in which the coefficient pair (ϕ_1, ϕ_2) lies close to the Fibonacci boundary landmark at $(1, 1)$. In the (ϕ_1, ϕ_2) plane, the Fibonacci point $(1, 1)$ sits at the boundary of the stable region and serves as a geometric reference: pairs whose dominant root $|r|$ falls in a narrow band below unity, consistent with the observed damped-oscillator behaviour in Table 1, define the admissible neighbourhood. The exact extent of this neighbourhood is determined by the division limit (Rule 3) and lifespan (Rule 4) constraints of Boman's tissue code, which together bound $|r|$ from above and below. A formal characterisation and visualisation of this region is provided in the companion platform at `https://par2discovery.com` (Root-Space Geometry page).

7. Testable Predictions and Experimental Designs

The PAR(2) framework is simple enough to support explicit, falsifiable predictions. For example, circadian disruption (e.g., Bmal1 loss or environmental misalignment) should increase variance in (ϕ_1, ϕ_2) across crypts and broaden the distribution of root magnitudes $|r|$ and phases $\arg(r)$, shifting the coefficient cloud away from a Fibonacci-consistent region [8]. Long-lived DCLK1$^+$ tuft cells should exhibit a damped AR(2)-like overshoot after injury, with longer return times (larger $|r|$) in chronic inflammatory contexts [10]. Chronotherapy that restores phase alignment should narrow dispersion in (ϕ_1, ϕ_2), shrink the spread of $|r|$, and decrease inter-crypt variance in tuft/DCS observables. Finally, agent-based simulations implementing Boman's five rules could be used to generate synthetic time-series of crypt observables and verify that stable Fibonacci-like regimes produce complex roots with $|r| < 1$ and coherent phase, while perturbations to individual rules map to distinct deformations in AR(2) coefficient space [2].

8. Discussion and Outlook

Boman's contributions to Fibonacci biology and tissue organization provide a powerful framework for understanding how simple rules can generate complex, ordered patterns in tissues [1, 2]. The Phase-Gated AR(2) model proposed here is intentionally modest: a low-dimensional time-domain representation whose parameters can be estimated from existing and future data, and which makes explicit contact with Fibonacci recurrences via its characteristic roots. The present work therefore sits at the intersection of Boman's tissue-renewal matrices [1, 2], the crypt-wide cellular and signalling landscape described by Nguyen, Lausten and Boman [6], and the Integrated Temporal-Spatial Oscillator Hypothesis for Crypt Renewal [11], specialising the latter to the specific case of colonic crypt dynamics and Fibonacci tissue codes.

In Boman's Fibonacci Quarterly model, asymmetric division and maturation rules are encoded in a low-dimensional transition matrix whose eigenvalues generate Fibonacci sequences in time and stable cell-type ratios [1]. In the present work, the proposed Phase-Gated AR(2)/VAR(2) framework plays an analogous role at the level of crypt observables: the AR/VAR coefficients form a companion matrix whose eigenvalues encode growth, decay and oscillation of renewal-related variables. Rather than demanding exact Fibonacci recurrences, Boman's solution is treated as a reference point in this eigenvalue space, and the question is whether homeostatic crypts lie near a "Fibonacci-consistent" region, with circadian disruption, injury or oncogenic mutation manifesting as systematic excursions away from that region.

For a journal like *The Fibonacci Quarterly*, the main message is that Fibonacci-like structure in tissues need not be confined to static spatial counts. It can also appear in the temporal structure of phase-aligned return dynamics, with golden-ratio-related constraints shaping the AR(2) coefficient space. Boman's five-rule tissue code can be viewed as defining a family of admissible PAR(2) models, with pathological states corresponding to excursions away from this family. The Fibonacci-consistent manifold has been defined above (Section 6) as the subset of the stable AR(2) stationarity triangle—the biologically admissible parameter space—in which the coefficient pair lies close to the Fibonacci boundary landmark at $(1, 1)$, with the admissible neighbourhood bounded by the division-limit and lifespan constraints of Rules 3 and 4. Future work should apply multivariate PAR(2) and VAR(2) models to circadian and injury time-series from crypts and organoids, and integrate spatial Boman-style simulations with time-domain summaries to build a unified spatio-temporal Fibonacci framework in which the manifold boundary can be validated against experimental perturbations.

9. Data Availability

The Phase-Gated AR(2) / temporal-spatial oscillator framework is archived on Zenodo under DOI 10.5281/zenodo.17366927 [11]. The organoid transcriptomic data reanalysed here are publicly available from the NCBI Gene Expression Omnibus under accession GSE157357 [9], as originally reported in Stokes et al. [8]. No additional

primary datasets were generated for this reply.

10. AI Use Declaration

In accordance with the Taylor & Francis AI Policy, the author declares the following. Generative AI tools were used during the preparation of this manuscript. Specifically, ChatGPT (model: GPT-5.1 Thinking, OpenAI, 2025) was employed to assist with: (i) editing and refining the English prose for clarity, concision and consistency; (ii) LaTeX and document structuring, equation typesetting, and consistency checks; (iii) support in formulating and refining the PAR(2)/VAR(2) mathematical framework, including notation, companion-matrix forms and explanatory text; and (iv) guidance on exploratory analysis of publicly available transcriptomic data (e.g., structuring time-series inputs, suggesting AR(2) fitting strategies and summarising root behaviour) based on datasets and results specified by the author. These tools were used as aids to drafting, calculation and exposition rather than as independent scientific authors. All scientific hypotheses, modelling choices, and formal conclusions presented in this article were reviewed, curated and integrated by the human author, who has critically assessed all AI-assisted content and takes full responsibility for the accuracy, integrity and originality of the work.

Disclosure of interests. The author has filed a UK patent application (GB2518973.9) covering aspects of the AR(2) eigenvalue methodology presented in this paper. No other conflicts of interest are declared.

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